

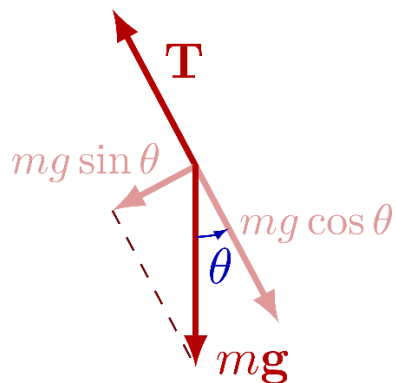
Measuring the Period of Oscillations of a pendulum

ENGS121 Mechanics Lab Section D

27 March, 2025

1. Introduction

The experiment is done to observe the period of a pendulum and determine whether human reaction time causes a significant error in timing measurements. The period of a simple pendulum is investigated to analyze random continuous data in means of measuring the period of oscillations.



The theoretical value of a period of oscillations is $T = 2\pi \sqrt{\frac{l}{g}}$.

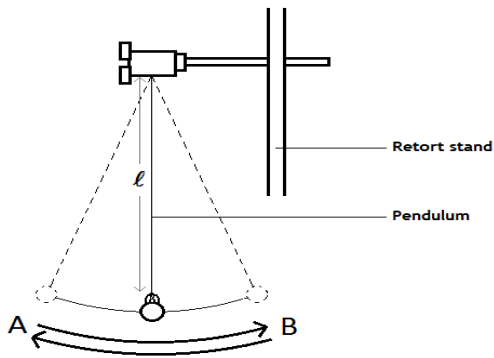
2. Measurements

Instruments

Name	Quantity	Properties
Ruler	1 pc	Wooden, 30.0 cm, resolution of 1 mm

Stand	1 pc	Stand
Thread	1pc	30.0 cm
Metal Ball (Tip of the Pendulum)	1pc	Metallic
Stopwatch	1 pc	Smartphone was used with 0.01 s resolution

Experiment Procedure



1. Build the simple pendulum by fixing the thread in the stand and attaching the ball on the tip of the thread, as shown in the diagram above.
2. The length of the thread to be measured and recorded.
3. Measure the period of 10 oscillations of the pendulum with the timer.
4. Repeat the previous step 60 times and record it in (Table 1).
5. Hold the ruler vertically with the bottom of the ruler level with the top of the subject's hand and experiment carried out to catch the ruler as fast as possible. Distance of the ruler drops before being caught to be recorded in (Table 3) indicating reaction time.

List of Variables

Variable	Value	Resolution
Period	Measured	0.01s
Length of the Thread	Controlled	1 mm
Amplitude	Controlled	1 mm

Raw Oscillation Data for 10 Oscillations (Table 1)

n	1	2	3	4	5	6	7	8	9	10
t (s)	11.88	11.04	11.63	11.36	11.62	11.36	11.69	11.52	11.82	11.67
n	11	12	13	14	15	16	17	18	19	20
t (s)	11.57	11.65	11.52	11.53	11.7	11.53	11.46	11.61	11.7	11.68
n	21	22	23	24	25	26	27	28	29	30
t (s)	11.55	11.23	11.4	11.48	11.51	11.24	11.5	11.56	11.46	11.71
n	31	32	33	34	35	36	37	38	39	40
t (s)	11.76	11.48	11.7	11.51	11.53	11.76	11.2	11.41	11.7	11.51
n	41	42	43	44	45	46	47	48	49	50
t (s)	11.35	11.58	11.61	11.43	11.48	11.61	11.56	11.43	11.58	11.23
n	51	52	53	54	55	56	57	58	59	60
t (s)	11.73	11.66	11.46	11.46	11.44	11.56	11.6	11.7	11.64	11.91

Oscillation Data for Single Oscillations (Table 2)

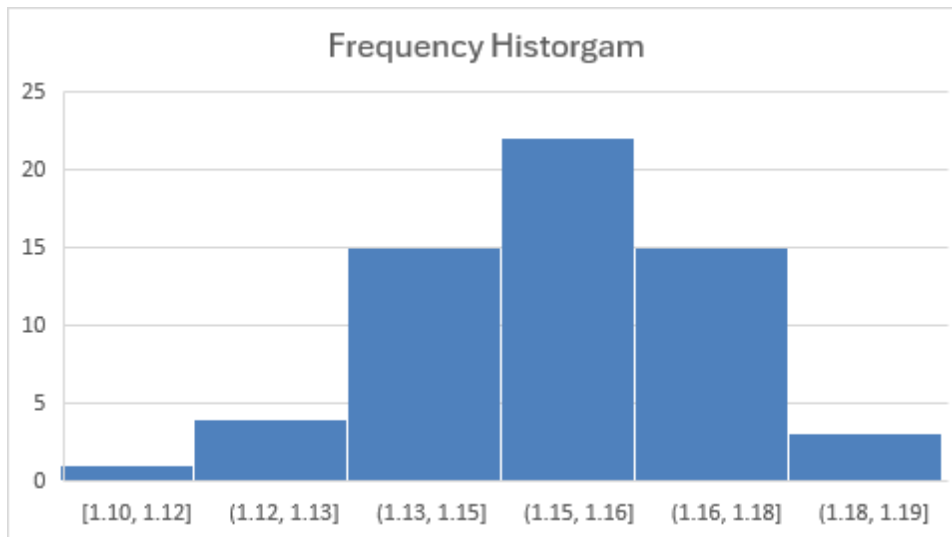
n	1	2	3	4	5	6	7	8	9	10
t (s)	1.19	1.10	1.16	1.14	1.16	1.14	1.17	1.15	1.18	1.17
n	11	12	13	14	15	16	17	18	19	20
t (s)	1.16	1.17	1.15	1.15	1.17	1.15	1.15	1.16	1.17	1.17
n	21	22	23	24	25	26	27	28	29	30
t (s)	1.16	1.12	1.14	1.15	1.15	1.12	1.15	1.16	1.15	1.17
n	31	32	33	34	35	36	37	38	39	40
t (s)	1.18	1.15	1.17	1.15	1.15	1.18	1.12	1.14	1.17	1.15
n	41	42	43	44	45	46	47	48	49	50
t (s)	1.14	1.16	1.16	1.14	1.15	1.16	1.16	1.14	1.16	1.12
n	51	52	53	54	55	56	57	58	59	60
t (s)	1.17	1.17	1.15	1.15	1.14	1.16	1.16	1.17	1.16	1.19

Errors

Name	Type	Solution
Reaction Time	Random	The number of measurements is to be increased in order to minimize the error. Timer to start when the pendulum has already completed one oscillation in order to minimize the reaction time delay.
Amplitude	Random	Fixing the starting point could help to always start from the same height and distance and minimize the displacement.

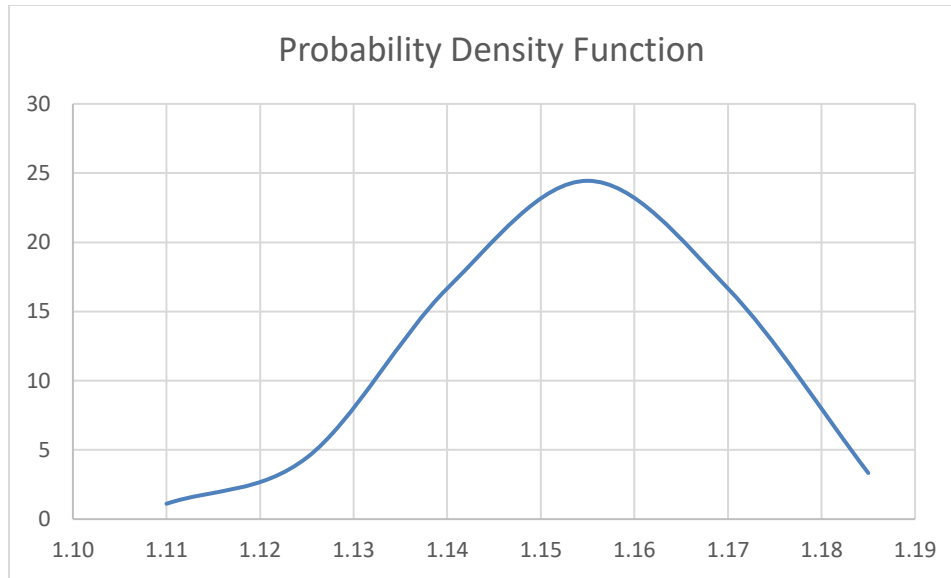
3. Data Analysis

The oscillation times were divided by 10, to get the time for single oscillations. Later frequency distribution and probability density distribution histograms were constructed, with bin size of 0.015 s.



Interval (s)	Midpoint (s)	Frequency	Probability	Probability Density
1.10-1.12	1.11	1	0.02	1.11
1.12-1.13	1.13	4	0.07	4.44
1.13-1.15	1.14	15	0.25	16.67
1.15-1.16	1.16	22	0.37	24.44
1.16-1.18	1.17	15	0.25	16.67
1.18-1.19	1.19	3	0.05	3.33

This Table is needed for constructing a Probability Density Function, where X is Midpoints and Y are Probability Density Values. Probability Density = Probability / Length of the Interval (bin size).



$$T_{mean} = \frac{\sum T_i}{N}$$

$$N = 60$$

$$\sum T_i = 69.28$$

$$T_{mean} = \frac{69.28}{60} = 1.15$$

$$T_{peak} = 1.16 \text{ (from the PDF)}$$

$T_{peak} \approx T_{mean}$ thus, the data is well-distributed and symmetrical.

Let us calculate the free fall acceleration from our measurements.

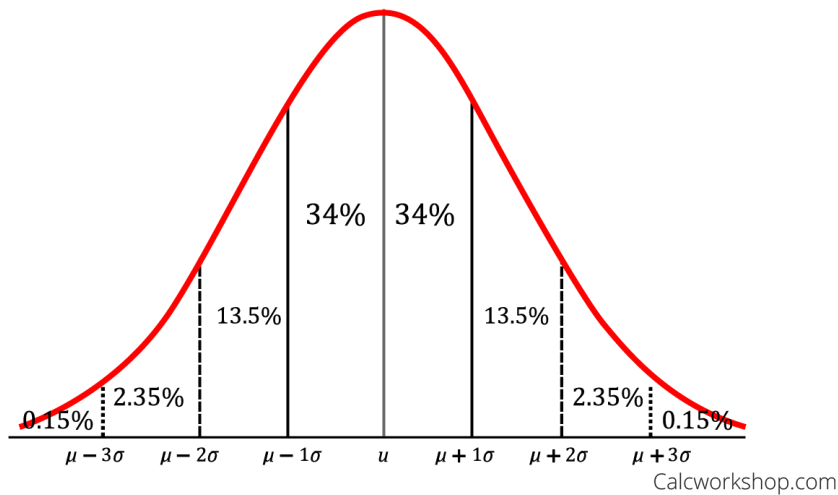
$$T = 2\pi\sqrt{\frac{l}{g}} \text{ thus, } g = 4\pi^2 \frac{l}{T^2} = 8.88 \text{ m/s}^2 \text{ while the real accepted value of } g \text{ is } 9.81 \text{ m/s}^2$$

The value of g that we measured from our calculations is 9.48% off from the real value of g .

$$\sigma = 0.02 \text{ (s)}$$

Empirical Rule (Confidence Intervals)

68%	95%	99.7%
$T_{mean} \pm \sigma$	$T_{mean} \pm 2\sigma$	$T_{mean} \pm 3\sigma$
$1.15 \pm 0.02 \text{ s}$	$1.15 \pm 0.04 \text{ s}$	$1.15 \pm 0.06 \text{ s}$



Empirical rule: 68% of data lies within $\mu \pm \sigma$, 95% within $\mu \pm 2\sigma$, and 99.7% within $\mu \pm 3\sigma$.

Ruler Drop Test

Supplementary Experiment (Table 3)

n	1	2	3	4	5	6
x cm	17.00 cm	13.00 cm	18.00 cm	20.00 cm	17.00 cm	16.00 cm

$$t = \sqrt{\frac{2x_{mean}}{g}}, \text{ where } x_{mean} = 16.83 \text{ cm, } g = 980 \text{ cm/s}^2$$

$t = 0.18 \text{ s}$ (Reaction Time)

The reaction time is much higher than the standard deviation, which suggests that we have a systematic error in our experiment, tied to the reaction time. Even though the data is relatively precise, the measured periods are closely clustered around the mean. Thus, Human reaction time errors bring more uncertainty than the experiment itself.

4. Conclusion and Evaluation

The experiment conducted investigates the period of a simple pendulum and the effect of reaction time on the measurements was elaborated. When comparing the value of gravitational constant that we got from the experiment to the approved value there is a 9.48% error. From the

measurements the average period of the pendulum with a 30.00 cm thread length turned out to be 1.15 s with a standard deviation of 0.02 (s).

5. References

Force diagram of a simple pendulum. (n.d.). TikZ.net. https://tikz.net/wp-content/uploads/2023/10/dynamics_pendulum-001.png

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